

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Solution Requirements

Online Complaint Registration System — MERN Stack Complaint
Management Platform

Phase: Brainstorming & Ideation Phase

Document Version: 1.1

Date: 18th of june, 2026

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Classification: Project Documentation — For Academic Submission

Solution Requirements

1. Introduction

This document specifies the complete set of Functional and Non-Functional Requirements for the Online Complaint Registration System. Each requirement is assigned a unique identifier to maintain traceability from requirement analysis through design, implementation, testing, and deployment.

The Online Complaint Registration System is a MERN-based full-stack web application designed to allow citizens to register complaints, track complaint progress, communicate with assigned agents, receive notifications, submit feedback, and allow administrators to manage users, agents, complaints, assignments, analytics, and workflow operations.

2. Functional Requirements

FR-01 to FR-05 – User Management & Authentication

Req. ID	Requirement Description	Priority	USN
FR-01	Register by providing name, email, password.	Must Have	USN-1
FR-02	Authenticate using email/password.	Must Have	USN-2
FR-03	Generate signed JWT token.	Must Have	USN-2
FR-04	Support RBAC (Citizen, Agent, Admin).	Must Have	USN-3
FR-05	Restrict unauthorized access.	Must Have	USN-3

FR-06 to FR-10 – Complaint Management

Req. ID	Requirement Description	Priority	USN
FR-06	Lodge complaint (title, category, desc, location).	Must Have	USN-4
FR-07	Store complaints in MongoDB Atlas.	Must Have	USN-4
FR-08	View own complaints.	Must Have	USN-5
FR-09	Display status (Submitted, Assigned, In Progress, Resolved).	Must Have	USN-5
FR-10	Show complaint details, timeline, and task management.	Must Have	USN-5

FR-11 to FR-15 – Agent Workflow

Req. ID	Requirement Description	Priority	USN
FR-11	Agent login requiring admin approval.	Must Have	USN-8
FR-12	View assigned complaints.	Must Have	USN-8
FR-13	Update complaint status.	Must Have	USN-9
FR-14	Communicate via chat.	Must Have	USN-10
FR-15	Restrict access to unassigned complaints.	Must Have	USN-8

FR-16 to FR-20 – Admin Operations

Req. ID	Requirement Description	Priority	USN
FR-16	View all users/agents.	Must Have	USN-13
FR-17	Approve/Reject agents.	Must Have	USN-11
FR-18	Assign complaints.	Must Have	USN-12
FR-19	Monitor complaints.	Must Have	USN-14
FR-20	View analytics.	Should Have	USN-14

FR-21 to FR-25 – Communication, Notifications & Feedback

Req. ID	Requirement Description	Priority	USN
FR-21	Socket.IO real-time chat.	Must Have	USN-5, USN-10
FR-22	Store chat messages.	Must Have	USN-5
FR-23	Notifications (Assignment, Status, Resolution).	Must Have	USN-6
FR-24	Submit feedback/ratings.	Should Have	USN-7
FR-25	View feedback.	Should Have	USN-15

3. Non-Functional Requirements

Req. ID	Category	Requirement Description	Priority
NFR-01	Security	Secure password hashing (Bcrypt).	Must Have

Req. ID	Category	Requirement Description	Priority
NFR-02	Security	JWT token authentication for API routes.	Must Have
NFR-03	Security	Role-based authorization (RBAC).	Must Have
NFR-04	Usability	Responsive UI for desktop/mobile.	Must Have
NFR-05	Usability	Mobile screen support.	Must Have
NFR-06	Performance	API response time < 500ms.	Should Have
NFR-07	Reliability	Complaint status consistency.	Must Have
NFR-08	Availability	Hosted via Vercel/Render.	Should Have
NFR-09	Scalability	MERN modular architecture.	Could Have
NFR-10	Maintainability	MVC (Routes/Controllers/Models).	Should Have
NFR-11	Data Integrity	Accurate MongoDB storage.	Must Have
NFR-12	Real-Time	Socket.IO implementation.	Must Have

4. Requirements Traceability Matrix

Req. ID	Design Artefact	Implementation File / Module	UAT Test Case
FR-01,02,03	Solution Arch.	authController.js, AuthContext.jsx	UAT-001,002,003
FR-04,05	RBAC Design	authMiddleware.js, ProtectedRoute.jsx	UAT-004,005
FR-06,07	DFD	complaintController.js, Complaint.js	UAT-006,007
FR-08,09,10	Dashboard Design	MyComplaints.jsx, ComplaintDetails.jsx	UAT-009,010,011
FR-11,12,13	Agent Workflow	AgentDashboard.jsx, agentController.js	UAT-015,016,017
FR-16,17,18	Admin Workflow	AdminDashboard.jsx, adminController.js	UAT-019,020,021
FR-20	Analytics Design	AdminAnalytics.jsx, analyticsController.js	UAT-023
FR-21,22	Chat Design	ChatPage.jsx, socketService.js	UAT-024,025,026
FR-23	Notif Design	NotificationBell.jsx, notificationController.js	UAT-028,029,030
FR-24,25	Feedback Design	FeedbackPage.jsx, feedbackController.js	UAT-032,034

5. Conclusion

The Solution Requirements document defines the functional and non-functional requirements necessary for the successful development and deployment of the Online Complaint Registration System. The functional layers map out all programmatic boundaries governing user management, ticket lifecycles, socket interaction, and analytics tracking.
